Module 1 Overview of Information Management

Learning Objectives

1. Know information from the perspective of individual;
2. Capitalize information as an important business resource;
3. Study information management as an important skill.

An information management course is essential to people who are practicing IT. It is but natural to know the ins and outs in handling data that could be beneficial to stakeholders of an organization.

Most of the people fail to realize we are already enslaved by information. Information is already critical when we make our decisions. Day-to-day, monthly, annually, five years from now --- you name it! All of the things we are planning to do must have a basis to come up with a wise decision. And the basis to that decision is information.



Figure 1-1 Information is the key.

But information is cheap. If you want to know today's weather or the week's span, you just wait for the news channel to flash it. Need to pay your bills? Use your phone and pay for it with your e-wallet. Catch the latest trends in technology, machine learning, gossip with anything. Go to youtube!

For you it is cheap! But organizations such as Google, Amazon, Facebook and other data-savvy companies consider all the information they gather from us as a gold mine. These companies can secure their future 5 years and beyond just knowing the behavioral information about you.

There goes a saying: think before you click!

As long companies know your profile and your behavior online, they can predict with a level of accuracy of the next decision you will make in the future. All the clicks, the visits you make online, registers to their information repository to record the pattern you are making. That is already an amazing feat! Would you believe they know more about your behavior compared to how you know yourself?



Figure 1-2 Companies analyzing behavioral patterns.

No worries, reality bites!

But instead of getting afraid with your time spent online, why not know these big companies on how they operate in recording the information they gather from us? What are their tools? How do they manage the data they collect from you? Is there an efficient way in storing/retrieving information?

You always say information. Why not data?

Well, why not ask about bits, bytes, records?

Information is much higher than data. Collected data is information. You make a wise decision based on the insights of information. Yes, you are correct. You can not generate a good ounce of information if you can not handle your data first.

As a person who is knowledgeable in information management, you are tasked to design a database that is efficient and true to itself in storing and retrieving data. Information management is an important skill even to technical people like us.

Information is a corporate resource. Insights derived from these tons of data are essential to keep organizations moving. It ensues businesses afloat in the investments they dive into for the sake of their firm. Companies are paying a good amount of money just to store and maintain it by people like us in the IT industry.

The race is on

Companies often realize the need to expand and improve software and hardware infrastructure is continuous. Multinational corporations must set-up data centers to meet the never ever ending demand for storage. It must be geographically located near to its customers. This requires huge investments to cope up with the demand.

With the cloud infrastructure in the picture, companies now opt to avail the services offered by tech giants to meet fluctuating computing needs. Multi-billion dollar e-commerce tech giants such as Amazon offer a wide variety of products that may be rented depending on the demand of the client.

Cloud computing is pay-as-you-go technology services that may include the following:

- Compute
- Storage
- Databases
- Networking
- Software
- And many more

With cloud computing, companies do not need to worry about huge investments in improving IT infrastructure. You can even scale down its computing power if there is a small demand for the access of your data. It is like renting supercomputers when there is a peak that demands it.

In terms of databases, products/services of tech giants that are competing for the business are:



Figure 1-3 Leading contenders in the business.

Of market share in the business, Amazon Web Services dominates the scene with 32.1%; SQL Azure 18.1%; Google Cloud Platform 6% as of this writing.

We can not discount the possibility that these leaders in the cloud computing business may be outranked by other players which are aggressive to have its slice in the pie. Companies like Alibaba, Huawei, and IBM are actively participating in developing cloud applications ready for use.



Figure 1-4 Other players in the cloud computing business.

Relational vs. non-relational

Relational databases dominate usage when we talk about structured data. Records here are organized by tables that consist of rows and columns. Tables can be connected (related) to other tables to form information. The structured way of organizing the data almost eliminates

redundancy because it is based on the mathematical concept of relational algebra. Its popular usage started in the 1970s and it is still today.

Structured Query Language (SQL) is a domain-specific language designed to maintain data for relational databases. SQL is sometimes pronounced sequel. Thanks to ANSI/ISO standard SQL, you can almost write similar query syntax without conflict across several database management systems (DBMS) vendors. Standard SQL SELECT script written in MySQL should run in Oracle, DB2, SQLServer without a problem. Obtaining a database server comes with a huge licensing price. DBMS softwares is proprietary and fully supported by vendors. Others opt to use open-source DBMS offered by PostgreSQL.



Figure 1-5 An open-source relational database.

Companies like IBM, Oracle, Microsoft are a few DBMS vendors that have been quite successful in the past and have their fair share in selling on-premise database servers.

On the other hand, a non-relational unstructured database offers a relaxed mode of data handling. Most of the data in the internet is unstructured so flexibility is one of the strengths in maintaining stored information. NoSQL is another term used to identify non-relational databases.

NoSQL is a document-oriented database that works with key-value stores and is normally encoded with javascript object notation (JSON). It does not follow the strict format of relational databases that work with schemas. NoSQL is considered a schema less database. It is far more superior in terms of scalability and is able to handle large volumes of data at much faster paced compared to relational databases.

MongoDB, Firebase, CouchDB, Cassandra, and among others are few of the popular NoSQL databases. NoSQL databases are commonly open-source and do not require a strict licensing of proprietary SQL softwares. As with the usual complaint for open-source softwares, it is often short or lacks support that proprietary software prides itself on.



Figure 1-6 Popular NoSQL databases.

The focus of our study will be more on relational databases. It is but most importantly to know the structure of relational databases that has existed for more than 50 years and it is still widely used by organizations today.

With the onset of cloud computing technology, these same companies are competing to bag the top rank in gaining market share. Cloud computing is a bundled service and database is one of the many offers that clients may avail as a shift from on-premise servers. Leading the way is Amazon AWS, Microsoft Azure, and Google Cloud Platform in that particular order of market shares.

Transactional vs. Analytical

Day-to-day sales are considered transactional in nature. As data in relational transactional databases becomes extremely valuable, analysts exert considerable effort to extract data for the study of trends and patterns.

There lies the difference between online transactional processing (OLTP) versus online analytical processing (OLAP). It has been argued that the two must have a separate server specific to its core functionality.

Transactional database serves as a frontline database that is used for day-to-day business of a company. Business processes may include sales, generation of invoices, inventory, reconciliation, and other accounting transactions.

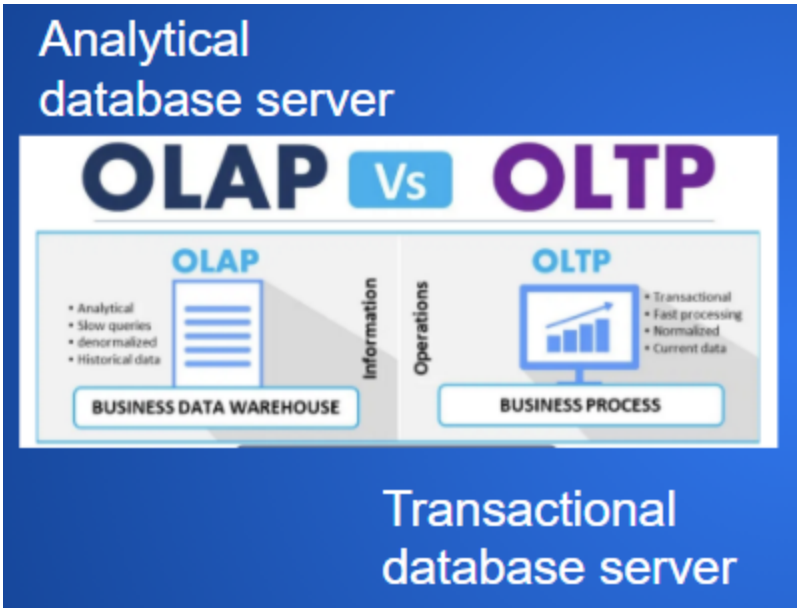


Figure 1-7. Transactional vs. Analytical database servers

Analytical databases are designed for analysis purposes in which one can derive patterns from the bulk of data a server possesses. Thus, the term data warehouse is coined. Data warehouse addresses the analytical processing needs of an organization. To put up a data warehouse, it is still dependent on transactional databases as its main source of data. It operates in a different architecture fit for analytical purposes.

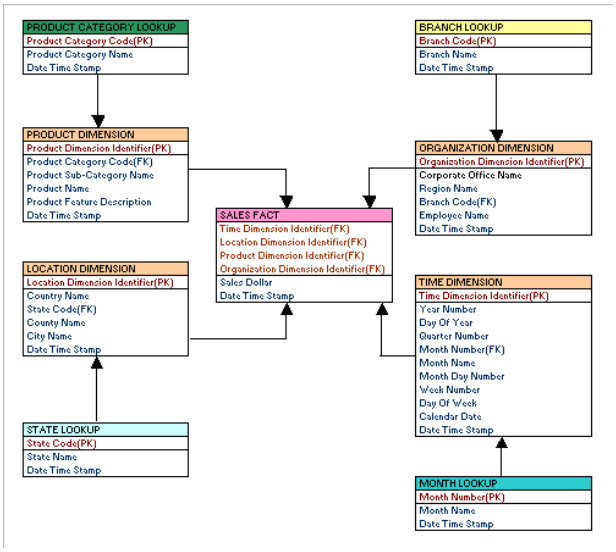


Figure 1-8. Data warehouse snowflake schema

Though the data warehouse is still a relational database, it is structured mostly in a snowflake schema. A great deal of effort is needed to maintain a data warehouse. This is where you hear the term extract-transform-load (ETL). ETL is basically a process of cleaning/segregating the

data (extract), repurposing/formatting the data that would fit in snowflake architecture (transform), and saving all the transformed data(load) to the server.

Data lakes

The demand for the insights from a data store continues to grow day-to-day as a survival mechanism for big companies today. Organizations do not rely anymore on one data source(transactional database) but with various sources coming from social media, web posts, video feeds. How to come up with valuable insights as quickly as possible is a continuous task to gain competitive advantage.

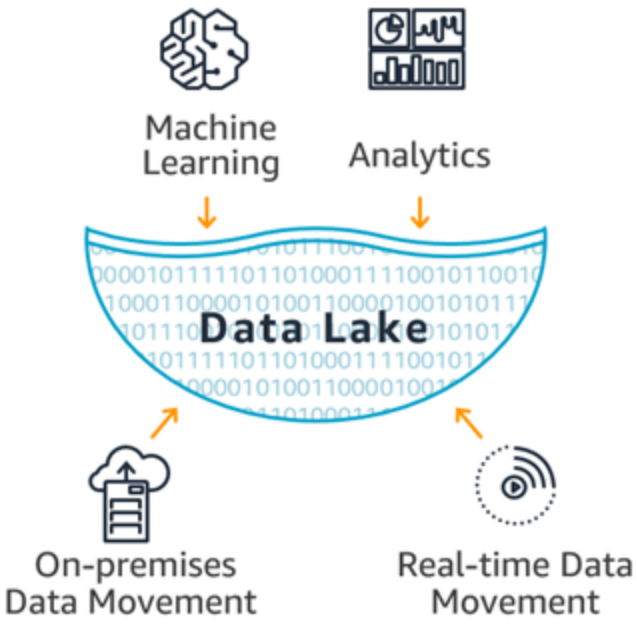


Figure 1-9. Data lake

From the definition of Amazon, a data lake is a centralized, curated, and secured repository that stores all your data, both in its original form and prepared for analysis. A data lake enables you to break down data silos and combine different types of analytics to gain insights and guide better business decisions.

The heavy lifting in the formation of data lakes is one of the services offered by Amazon Web Services cloud computing technology.

With the on-set of machine learning and demand for more analytics, organizations depend on its people who have the skills to grasp today's technologies going forward. It is important for an IT specialist to decipher the different techniques that will help the organization to attain its mission through insightful information and come up with better and sound decisions.

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